

## 4.4.2 Atoms and nuclear radiation

### 4.4.2.1 Radioactive decay and nuclear radiation

| Content                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Key opportunities for skills development |
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| <p>Some atomic nuclei are unstable. The nucleus gives out radiation as it changes to become more stable. This is a random process called radioactive decay.</p> <p>Activity is the rate at which a source of unstable nuclei decays.</p> <p>Activity is measured in becquerel (Bq)</p> <p>Count-rate is the number of decays recorded each second by a detector (eg Geiger-Muller tube).</p> <p>The nuclear radiation emitted may be:</p> <ul style="list-style-type: none"><li>• an alpha particle (<math>\alpha</math>) – this consists of two neutrons and two protons, it is the same as a helium nucleus</li><li>• a beta particle (<math>\beta</math>) – a high speed electron ejected from the nucleus as a neutron turns into a proton</li><li>• a gamma ray (<math>\gamma</math>) – electromagnetic radiation from the nucleus</li><li>• a neutron (n).</li></ul> |                                          |
| <p>Required knowledge of the properties of alpha particles, beta particles and gamma rays is limited to their penetration through materials, their range in air and ionising power.</p> <p>Students should be able to apply their knowledge to the uses of radiation and evaluate the best sources of radiation to use in a given situation.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | WS 1.4, 1.5                              |

#### 4.4.2.2 Nuclear equations

| Content                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Key opportunities for skills development |
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| <p>Nuclear equations are used to represent radioactive decay.</p> <p>In a nuclear equation an alpha particle may be represented by the symbol:</p> ${}^4_2\text{He}$ <p>and a beta particle by the symbol:</p> ${}^0_{-1}\text{e}$ <p>The emission of the different types of nuclear radiation may cause a change in the mass and /or the charge of the nucleus. For example:</p> ${}^{219}_{86}\text{radon} \longrightarrow {}^{215}_{84}\text{polonium} + {}^4_2\text{He}$ <p>So alpha decay causes both the mass and charge of the nucleus to decrease.</p> ${}^{14}_6\text{carbon} \longrightarrow {}^{14}_7\text{nitrogen} + {}^0_{-1}\text{e}$ <p>So beta decay does not cause the mass of the nucleus to change but does cause the charge of the nucleus to increase.</p> <p>Students are not required to recall these two examples.</p> <p>Students should be able to use the names and symbols of common nuclei and particles to write balanced equations that show single alpha (<math>\alpha</math>) and beta (<math>\beta</math>) decay. This is limited to balancing the atomic numbers and mass numbers. The identification of daughter elements from such decays is not required.</p> <p>The emission of a gamma ray does not cause the mass or the charge of the nucleus to change.</p> | <p>WS 1.2, 4.1</p> <p>MS 1b, c, 3c</p>   |

#### 4.4.2.3 Half-lives and the random nature of radioactive decay

| Content                                                                                                                                                                                                                                                                                            | Key opportunities for skills development |
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| <p>Radioactive decay is random.</p> <p>The half-life of a radioactive isotope is the time it takes for the number of nuclei of the isotope in a sample to halve, or the time it takes for the count rate (or activity) from a sample containing the isotope to fall to half its initial level.</p> |                                          |

| Content                                                                                                                                             | Key opportunities for skills development |
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| Students should be able to explain the concept of half-life and how it is related to the random nature of radioactive decay.                        | WS 1.2                                   |
| Students should be able to determine the half-life of a radioactive isotope from given information.                                                 | MS 4a                                    |
| (HT only) Students should be able to calculate the net decline, expressed as a ratio, in a radioactive emission after a given number of half-lives. | (HT only) MS 1c, 3d                      |

#### 4.4.2.4 Radioactive contamination

| Content                                                                                                                                                                                                                                                                                                                                                                                    | Key opportunities for skills development |
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| Radioactive contamination is the unwanted presence of materials containing radioactive atoms on other materials. The hazard from contamination is due to the decay of the contaminating atoms. The type of radiation emitted affects the level of hazard.<br><br>Irradiation is the process of exposing an object to nuclear radiation. The irradiated object does not become radioactive. | WS 1.5                                   |
| Students should be able to compare the hazards associated with contamination and irradiation.                                                                                                                                                                                                                                                                                              | WS 1.5                                   |
| Suitable precautions must be taken to protect against any hazard that the radioactive source used in the process of irradiation may present.                                                                                                                                                                                                                                               | WS 1.5                                   |
| Students should understand that it is important for the findings of studies into the effects of radiation on humans to be published and shared with other scientists so that the findings can be checked by peer review.                                                                                                                                                                   | WS 1.6                                   |